

Angad Basandrai

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EDUCATION

IIST Shibpur

Bachelor of Technology in Computer Science and Technology

Howrah, West Bengal

Aug. 2025 – Aug. 2029

RESEARCH

Safe Opponent Exploitation in Colonel Blotto | *cs.GT* | [github](#)

sep. 2026 – present

- implemented a double oracle framework with dynamic programming best-response oracles and linear programming over the equilibrium polytope to identify safety-constrained exploitation strategies.
- developed an exact solver for Colonel Blotto games containing up to 4.3×10^{12} **pure strategies** without enumerating the complete strategy space.
- characterised gift regions for unequal battlefield values and improved the equal-value divisibility condition from $n \mid B$ to $2B/n \in \mathbb{Z}$.
- evaluated exploitation strategies against **4,510 human allocations** from FiveThirtyEight Riddler Nation tournaments.

PROJECTS

qirc | *rust, quantum computing* | [github](#)

sep. 2026 – present

- implemented a qir compiler and built-in state-vector simulator from scratch in rust, supporting circuit validation, optimization, and execution.
- implemented multi-format output, including openqasm 3, qir, json, and hardware-specific gate-set and qubit-layout transformations.
- selected as a finalist in **SegFault 2026**, with an in-person presentation at IISc Bengaluru.

syrinx | *rust, opengl* | [github](#)

july 2026

- implemented a from-scratch browser engine in rust, integrating **6 subsystems** spanning html parsing, dom representation, layout, graphics, utilities, and browser chrome.
- built a rendering pipeline that converts parsed dom trees into layout boxes and display commands before rasterising pages through an opengl renderer.
- implemented **4 browser interactions** url navigation, hyperlink resolution, page reload, and home-page.

2d physics engine | *c++, opengl, glfw* | [github](#)

sep. 2025 – july 2026

- implemented a real-time 2d physics engine using c++, opengl, and glfw, with rk4 numerical integration.
- optimised the simulation pipeline to support **1,000+ bodies at 60+ fps** during real-time execution.
- implemented collision detection and physics-based interactions for simulated bodies.

tune-ai | *python, peft, reinforcement learning* | [github](#)

june 2026

- implemented a ppo-based prompt optimiser using qwen2.5-1.5b-instruct with lora fine-tuning and a groq-hosted llama 3.1 8b reward model.
- trained the optimiser on synthetic noisy restaurant-order examples to improve prompt robustness.
- used clipped policy and value losses with entropy regularisation and token-level advantage estimation.

py-scorch | *python, cuda* | [github](#)

may 2026 – june 2026

- implemented a neural network library in python with cuda support for gpu-accelerated computation.
- trained a neural network on mnist, achieving approximately 98% accuracy, comparable to pytorch while taking roughly twice the execution time.

TECHNICAL SKILLS

Languages: Rust, C++, C, Python, Java, JavaScript, SQL

Systems & Frameworks: LLVM, OpenGL, GLFW, React, PyTorch, PEFT

Tools: Git, Linux, Ghidra, Burp Suite

OTHER

Machine Learning Specialization from Stanford University and DeepLearning.ai

Rated 1000+ on Codeforces.com

Founder and maintainer of [iist.wiki](#), including its associated subwebsites